

Statistical Modeling & Inferencing

Assignment 1

Analysis of California Housing Prices

Student Name: Chennupati Sree Nihitha

Roll Number: 2025EM1100344

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Google Colab Link : [California Housing.ipynb](#)

Summary:

For this assignment, I chose to analyze the California Housing dataset (1990 Census data) to understand what factors actually drive property values. The dataset covers over 20,000 districts, offering a mix of location data, economic indicators, and housing structural details. My main goal was to build a regression model that could predict median house prices, but I also wanted to dig into the data to see *why* some areas are more expensive than others.

The process wasn't straightforward. I ran into significant issues with "multicollinearity"—basically, several variables were repeating the same information (like total rooms vs. population). To fix this, I had to engineer new features that measured density rather than just size.

In the end, I compared a standard Linear Regression model against a Ridge Regression model. My analysis found that Ridge Regression was the better choice because it handled the correlations much better. The biggest takeaway? Income is king. Neighborhood wealth was by far the strongest predictor of price, followed closely by whether the house was near the ocean or inland.

Part 1: Exploring the Data

Initial Observations When I first loaded the data, I looked at the distributions of the variables. The target variable, `median_house_value`, had a weird spike at the \$500,001 mark. It looks like the census takers "capped" any house worth more than that at \$500k. This is a bit of a problem because it means my model is going to have a hard time predicting the value of luxury homes accurately.

The Correlation Problem I ran a correlation matrix to see which variables moved together. Two things stood out:

1. Income predicts price: The `median_income` variable had a strong 0.68 correlation with house value. This makes sense economically.
2. Redundant features: I noticed that `total_rooms`, `total_bedrooms`, and `population` all had correlations over 0.9 with each other. This is logical—blocks with more people naturally have more rooms—but it's bad for modeling. It creates "noise" without adding new information.

Cleaning and Preprocessing To handle the data quality issues, I made a few key decisions:

- **Missing Data:** The `total_bedrooms` column had some gaps. I filled these in using the median value (instead of the mean) because the data was skewed, and I didn't want outliers throwing off the number.
- **Feature Engineering:** To solve the correlation problem I mentioned above, I created ratio variables. Instead of using raw "total rooms," I calculated `rooms_per_household`. This tells us how spacious the homes are, which is a much better indicator of value than just the size of the census block.
- **Categorical Data:** The `ocean_proximity` column was text, so I converted it into numbers using One-Hot Encoding.

Part 2: Building the Model

Dealing with Multicollinearity Before fitting the models, I checked the Variance Inflation Factor (VIF). Initially, the raw counts (like Population) had huge VIF scores, confirming they were problematic. However, after swapping them out for the ratio features I created in Part 1, the VIF scores dropped significantly. This gave me confidence to move forward.

Model Comparison I decided to pit two models against each other:

1. **Linear Regression (OLS):** The standard baseline approach.

2. Ridge Regression: A more advanced version that adds a penalty to shrink the coefficients. I chose this specifically because I was still worried about residual correlation between the features.

Results Both models actually performed quite similarly, with an R^2 of around 0.61. This means my model explains about 61% of the variation in housing prices. However, I chose Ridge Regression as the winner. Even though the accuracy was similar, Ridge is "safer"—it prevents any single feature from having an unrealistically large impact on the prediction, which makes the model more stable for future data.

Diagnostics I plotted the residuals (the difference between what my model predicted and the actual price).

- The Good: The errors mostly followed a bell curve (Normal distribution), which is what we want to see.
- The Bad: You can clearly see a line of errors corresponding to that \$500k price cap. My model kept predicting values higher than \$500k, but the actual data was stuck at the limit.

Part 3: Insights and Conclusion

What Drives Price? After looking at the model coefficients, a few clear patterns emerged:

1. Wealth: As expected, median income is the heavy hitter. If you know the income of a neighborhood, you have a very good guess at the house prices.
2. The "Inland Discount": The location data was fascinating. Houses marked as "INLAND" had a massive negative impact on price compared to coastal homes. It confirms that in California, you pay a premium for the ocean breeze.
3. Density: Areas with higher **population_per_household** tended to have lower values. This suggests that overcrowding is a negative factor for property value.

Real-World Limitations:

I have to acknowledge that this analysis has limits. The \$500k cap is the biggest one—it effectively blinds the model to the true value of high-end real estate. Also, since this data is from 1990, the actual dollar amounts are totally outdated (you definitely can't buy a house in the Bay Area for \$200k anymore!). However, the *relationships* hold true: location and income are still the primary drivers today.

Recommendations If I were advising a real estate investor based on this model, I'd say: Look for areas where median income is trending up, but the prices haven't caught up yet. And be wary of inland properties—unless there is a significant infrastructure change, they structurally command lower prices than the coast. For a future analysis, I'd definitely want to bring in modern data like interest rates or distance to tech hubs (like Silicon Valley) to make the predictions more relevant to 2025.